



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

1.0 Big Picture Overview

July 2025

The PETRONAS Group adopts zero tolerance against all forms of bribery and corruption. We abide by the PETRONAS Code of Conduct and Business Ethics (CoBE) & Anti-Bribery and Corruption (ABC) Manual, guided by our Shared Values and Statement of Purpose.

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Records of Change

Date of Issue/ Reissue	Revision No.	Reason of Development / Change
July 2025	00	New module offering inline with TI&R 2025.

Learning Outcome

At the end of the session, you will be able to:

- DESCRIBE the concepts of IIoT, Digital Twin, Industrial wireless, 5G, AI, ML, DL, RAO, Blockchain
- EXPLAIN the fundamentals, architecture, standards, and terminologies of IIoT, Digital Twin, Industrial wireless, 5G, AI, ML, DL, RAO, Blockchain
- DESCRIBE the use cases based on subscribe technologies.
- RELATE pacing and emerging technologies with industry 4.0 applications such as AI, IIoT, cloud computing, Autonomous Robot, cyber security, big data analytics, simulation and system integration.

Agenda

Time	Day 1	Day 2	Day 3	Day 4
AM	Introduction to SKG14 IC8 Programme Ice Breaking Integrity Moment Pre-Test	Topic 8: Introduction to Digital Twin	Topic 15: Why AI	Topic 18: What are the Machine Learning (ML) & Deep Learning (DL)
	Topic 1: Big Picture Overview of Industrial Internet of Things (IIoT), Digital Twin, Pervasive Wireless, Artificial Intelligence and Machine Learning.	Topic 9: Digital Twin Architecture	Topic 16: What is AI	Topic 19: Hardware Requirement for Computing AI, ML and DL
	Topic 2: Introduction to Industrial Internet of Things (IIoT)	Topic 10: Data Communication in Digital Twin	Topic 17: Data Analysis	Topic 20: Application of AI,ML and DL in PETRONAS in supporting Sustainability
	Topic 3:Key Technologies for IIoT	Topic 11: Digital Twin Applications		#Case Study 3: Hands-on Using Texas Wind Turbine Dataset (Sustainability)
	Topic 4: Industrial Internet of Things (IIoT) Reference Architecture	Topic 12: Industrial Wireless		
Lunch Break				
PM	Topic 5: IIoT Use Cases	Topic 13: Introduction to 5G	#Case Study 1: Hands-on Regression Techniques Using Datasets	#Case Study 4: Hands-on Using Pressure Sensor Dataset
	Topic 6: Hands-on Activities	Topic 14: Remote Operation & Maintenance	#Case Study 2: Hands-on Classification Techniques Using Datasets	Topic 21: Blockchain
	Topic 7: Reference, Standards and Further Reading			Post Test

Ground Rule

Have **FUN** with learning.

Respect others' point of view.

Open **communication**.

Learn from each other.

Come **prepared** to learn.

Observe **punctuality**.

There are **NO DUMB QUESTIONS**.



Ice-Breaking / Introduction



Name

Years in PETRONAS

OPUs in PETRONAS

Current Position

Experience with Metering System

Expectations of the course

Integrity Moment

[\[Open\]](#)

1.

Storytelling

1. Choose any topic that is relevant to integrity that you want to talk about in the span of 3-5 minutes.
2. You can share your own experience or others.
3. Highlight what is the lesson learnt.
4. Ask participants their opinion on the topic shared.

2.

Video sharing

1. Share a 2-3 minutes video on integrity related topic.
2. Here is a repository of integrity videos for you to choose from to facilitate the integrity moment [myPETRONAS](#).
3. You are allowed to used videos from external source but please observe PETRONAS sensitivity.
4. Ask participants their opinion on the video shared.

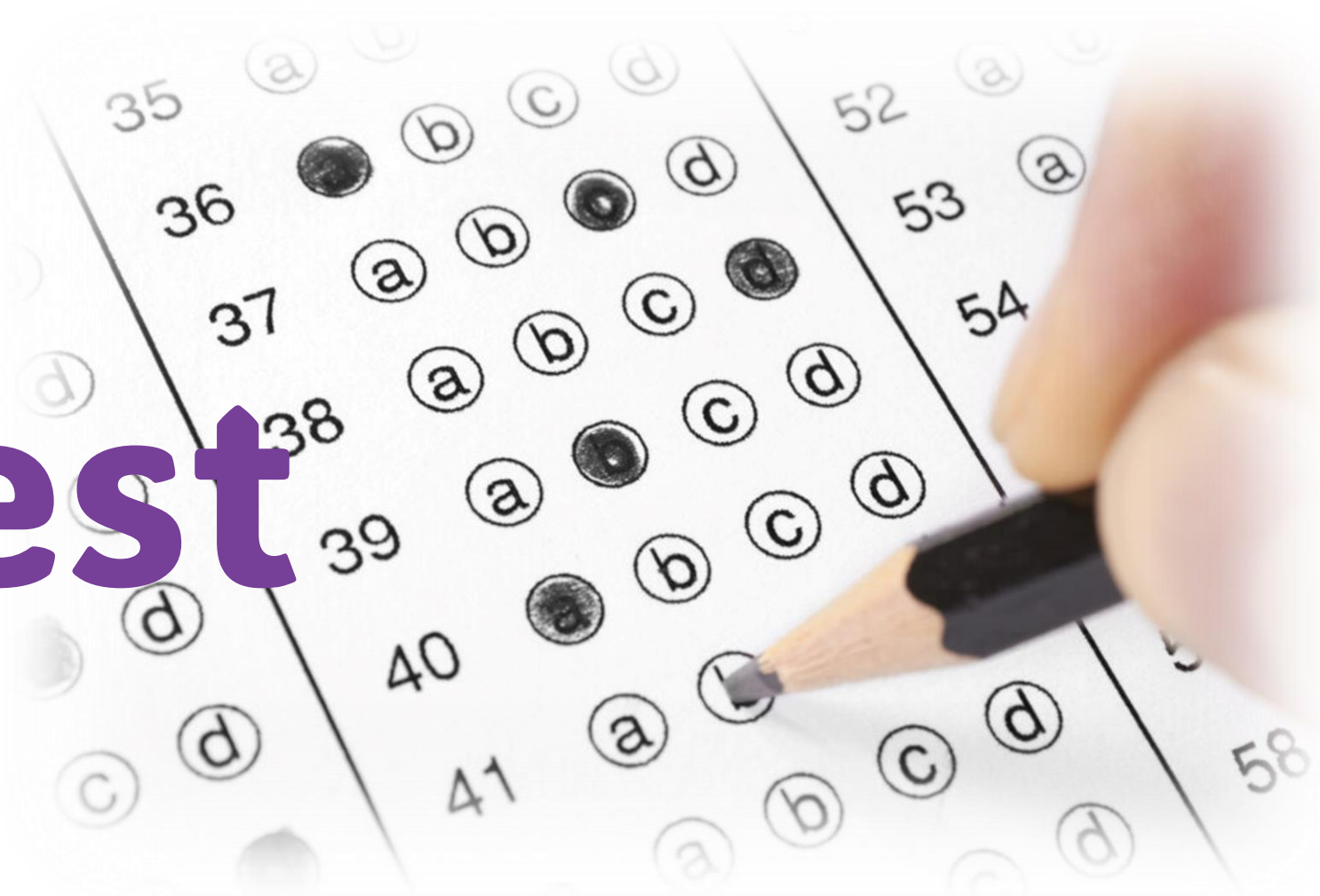
3.

Presentation

1. Present on any topics of your choice on integrity.

Assessment

Pre-Test



1.0 Big Picture Overview:

**Industrial Internet of Things (IIoT), Digital Twin,
Pervasive Wireless, Artificial Intelligence, Machine
Learning, RAO & Blockchain**

SKG 14 IC8

Pacing & Emerging Technologies Level 2

START

Click the START button to begin the course

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